Ahmet Safak Senel

Data Scientist & Computer Engineer

Istanbul, Turkey | safaksenel61@gmail.com | Live Portfolio | github.com/safaksenel | LinkedIn
DataCamp Certified (DEA) | Coursera / Google Certified (Data Science)

PROFESSIONAL SUMMARY

Computer Engineering graduate from Nisandasi University, honing low-level systems programming and algorithmic depth in C & Python within the Ecole 42 Istanbul Common Core curriculum. At FLO Group, I optimized 22 Turkish BERT language models using Intel OpenVINO, accelerating customer review triage by 85%. As a Google AI & Technology Academy (YZTA) graduate, I served as Product Owner & System Architect for BlindHire, designing bias-free Conversational AI and RAG architectures. Working across PyTorch, Scikit-learn, LangGraph, DuckDB, FastAPI, and Next.js, I don't just train models—I productionize them into sub-millisecond REST APIs and scalable AI systems.

WORK EXPERIENCE

Business Intelligence & Big Data Intern
03/2025 – 07/2025

FLO Group | Istanbul, Turkey

- Developed and trained **22 dual-tier Turkish BERT language models** classifying **10,000+ monthly customer reviews** across 11 root causes.
- Optimized inference pipelines with **Intel OpenVINO**, achieving a **4x CPU speedup**; enriched dataset using LLMs (ChatGPT, Gemini, Grok, DeepSeek) and analyzed misclassifications via SHAP, boosting weighted **F1 score from 0.72 to 0.892**.
- Engineered a Tkinter desktop GUI packaged as an executable (.exe) using PyInstaller for operational teams, reducing manual analysis time by **85% and saving 300+ man-hours monthly**.

`Python 3.14` `BERT / HuggingFace` `PyTorch` `Intel OpenVINO` `SHAP` `Tkinter GUI` `PyInstaller`

Data Science & AI Fellow / Team Lead
01/2026 – 10/2026

Google AI and Technology Academy (YZTA) | Istanbul, Turkey

- Graduated upon completing Bootcamp, 10/10 Coursera courses, and 17/17 Moodle modules; specialized in Deep Learning, NLP architectures, FastAPI microservices, Google BigQuery & SQL, and Agile/Scrum delivery.
- Engineered **Smart File Organizer v2.0** during a 72-hour hackathon—an event-driven filesystem daemon powered by Watchdog with unicode sanitization and GUI desktop packaging.
- Contributed to the academy's open-source Slack bot (Cemil) and served as Product Owner / Scrum Master delivering collaborative AI solutions in cross-functional teams.

`Python 3.14` `FastAPI` `Watchdog` `BigQuery / SQL` `Tableau` `Git / GitHub` `Agile / Scrum`

Library Assistant
09/2023 – 03/2025

Nisantasi University | Istanbul, Turkey

- Managed cataloging, indexing, and Excel-based database organization for over 10,000 academic assets.
- Optimized internal digital tracking and data organization, streamlining library inventory operations.

Prior Internships (DenizBank & Isik Schools)
2017 – 2023

DenizBank (Intern, 12/2022 – 01/2023) | Isik Schools (IT Intern, 09/2017 – 06/2018)

- **DenizBank:** Gained hands-on knowledge in digital banking operations, financial data workflows, and remote work discipline.
- **Isik Schools:** Supported local area network (LAN) administration, hardware maintenance, and troubleshooting.

EDUCATION

Ecole 42 Istanbul
08/2026 – Present

Software Engineering (Common Core Curriculum)

Admitted to the core curriculum following Piscine completion. Intensive peer-to-peer programming in C and Python focusing on dynamic memory management, low-level data structures, algorithmic complexity, and Unix system programming.

Google AI and Technology Academy (YZTA)
01/2026 – 10/2026

Data Science & Artificial Intelligence Program (Fellow & Graduate)

Nisantasi University
2023 – 2025

Bachelor of Science in Computer Engineering

Bahcesehir University
2022 – 2023

English Preparatory Program

Nisantasi University
2018 – 2020

Associate Degree in Computer Programming

FEATURED ENGINEERING PROJECTS

BlindHire: AI-Powered Anonymous Voice Interview & Assessment Platform
YZTA Capstone Project

Role: Product Owner & System Architect (5-Person Team) | GitHub • Live Demo

- Architected a **100% bias-free (DEI-compliant)** duplex conversational AI interview agent; integrated Groq (Llama 3.3) and LangChain RAG with **<400ms latency** for hallucination-free technical questions and natural interruption handling.
- Analyzed speech pace via Web Audio API; coordinated mathematical lip-sync avatar rendering, computer vision eye-tracking, and automated JSON scorecard evaluation (CV scoring) on FastAPI & Next.js.

FastAPI | LangChain & RAG | Groq (Llama 3.3) | Web Audio API | Computer Vision | Next.js & TypeScript | Docker

Agentic Copilot: Multi-Agent Data Science & Sandboxed Code Interpreter
Autonomous Data Analyst

github.com/safaksenel/agentic-datascience-copilot

- Constructed an autonomous data analysis graph using LangGraph state machines coordinating Query Router, SQL Generator, DuckDB Executor, AST Code Validator, and Explainer agents.
- Achieved **3.42ms analytical query speed over 500K+ rows using DuckDB in-memory OLAP**; engineered an AST Node Visitor sandbox blocking unsafe syscalls alongside self-healing reflection loops.

Python 3.14 | LangGraph | DuckDB OLAP | FastAPI | Python AST Sandbox | Pydantic v2 | Docker

Edge Vision Analytics: Real-Time Computer Vision & Edge Telemetry Pipeline
Edge Analytics

github.com/safaksenel/edge-vision-analytics-pipeline

- Engineered **2-stage ByteTRACK IoU object tracking**, ray-casting polygon geofencing intrusion detection, and EMA speed estimation for live camera feeds.
- Delivered **30+ FPS edge streaming** with **<8ms tracking loop latency** and 20 Hz WebSocket telemetry via FastAPI without requiring heavy cloud GPU infrastructure.

Computer Vision | ByteTRACK | FastAPI | WebSockets | OpenCV | Ray-Casting | Docker

YZTA 5.0 Datathon: Career Success Prediction & Telco Churn MLOps
Ensemble Machine Learning & MLOps

Datathon Project • Telco Churn API

- **YZTA 5.0 Datathon:** Extracted NLP signals from free-form mentor feedback via TF-IDF & SVD, blended Optuna-tuned LightGBM and CatBoost using Level-2 RidgeCV Stacking Ensemble to reach **0.1418 RMSE** across 56,000 rows.
- **Telco Churn MLOps:** Benchmarked 10 ML models, balanced data with SMOTE, and deployed a Soft Voting Ensemble (**0.8422 ROC-AUC**) on Docker via FastAPI handling 50k records/min with <25ms latency.

LightGBM | CatBoost | Stacking Ensemble | Optuna | TF-IDF NLP | Scikit-learn | FastAPI | Docker

TECHNICAL SKILLS

Languages: Python (3.14), SQL, C, TypeScript, R, Bash

Machine Learning & AI: PyTorch, Scikit-learn, HuggingFace Transformers (BERT), LightGBM, CatBoost, XGBoost, Optuna Bayesian Optimization, Intel OpenVINO, SHAP (XAI)

LLM, Agents & NLP: LangChain, LangGraph (Multi-Agent Systems), RAG, Groq (Llama 3.3), Prompt Engineering (CoT, Few-Shot), Natural Language Processing (NLP)

Computer Vision: OpenCV, ByteTRACK Object Tracking, Polygon Geofencing (Ray-Casting), EMA Speed Analytics

Data Analytics & OLAP: DuckDB (In-Memory OLAP), Pandas, NumPy, Google BigQuery, Tableau, Power BI, Statistical Modeling

Backend & MLOps: FastAPI, Django 6.1, Django Ninja, Pydantic v2, RESTful APIs, Docker, WebSockets, PostgreSQL (Supabase Port 6543), Redis, PyInstaller

Web & Automation: Next.js 16 (React 19), Tailwind CSS v4, Motion, Playwright, Crawl4AI, Selenium 4, Watchdog, Git / GitHub, Linux

OFFICIAL CERTIFICATIONS & ACCREDITATIONS

- **DataCamp Certified: Data Engineer Associate (DEA)**
  Verification ID: DEA0011477137123 (Feb 2026)
- **Google Data Analytics Professional Certificate**
  Google / Coursera (SQL, R, Tableau, Data Wrangling)
- **100 Days of Code: Python Pro Bootcamp (Udemy)**
  Python OOP, Selenium 4, Web Scraping, REST APIs
- **Google Project Management (Agile & Scrum) Certificate**
  Google / Coursera (Sprint Planning, Scrum Master)
- **Coursera / Google: Data Science Professional Badge**
  Verification ID: wvTvEoBsQ6u07xKAblOrdQ (Jul 18, 2026 • 11 Courses)
- **DataCamp Data Scientist with Python Career Track**
  Pandas, NumPy, Scikit-learn, Statistical Validation
- **Google AI & Prompting Essentials Specializations**
  Google / Coursera (LLM Prompting, CoT, Generative AI)
- **BTK Academy Data Science & Machine Learning Certificates**
  BTK Academy & Ministry of Industry

LANGUAGES

Turkish: Native

English: Advanced (C1) — Bahcesehir University English Prep

Spanish: Elementary (A1)

REFERENCES

Emre Guleryuz — BI & Big Data Manager, FLO Group
(Contact details available upon request)

Fevziye Adamcil Kucukay — Library Director, Nisantasi University
(Contact details available upon request)